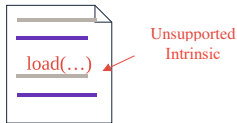
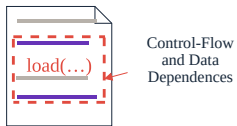


(a) Source Analysis

Locate Intrinsic

Control-Flow and
Data-Flow Analysis

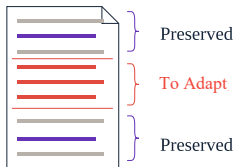
Bound Scope



Control-Flow and Data Dependences

Dependence Closure

Partition Regions



(b) Hierarchical Mapping

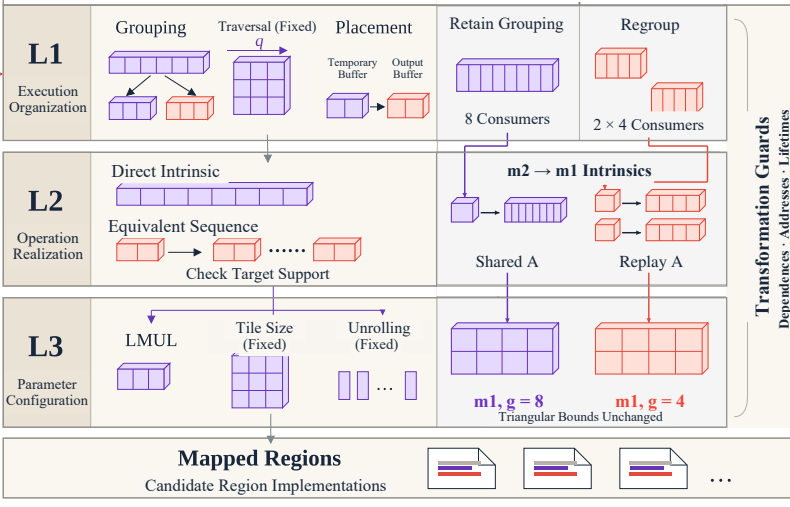


Target Constraints

Instructions · Registers · Memory Access

Mapping Decisions

Example: TRMM

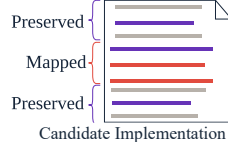
$$\text{RU kernel} \cdot \text{LMUL} = 1$$


Preserved Structure

Adapted Structure

(c) Adaptive Selection

Assemble Kernel



Target Compiler

RVV Source $\rightarrow$ Object Code

Validate & Measure

- ✓ ISA Checks
- ✓ Numerical Checks
- ⌚ Latency

Autotuning

Kernel Selection



c1	c2
	Lowest Measured Latency Among Qualified Candidates